Chapter summary

Familiarising yourself with **items in green** is important for effective revision.
Paying attention to items in the boxes is helpful for understanding the concept(s).

10.1 Price Elasticity of Demand

Meaning of price elasticity of demand

- **Price elasticity of demand** measures how responsive the quantity demanded of a good is to a change in its price.

$$\text{Price elasticity of demand coefficient } (E_d) = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

Even when absolute changes in the price and quantity demanded of two goods are the same, their price elasticities of demand may be different.

- According to the law of demand, the price elasticity of demand coefficient (E_d) is always negative.
- The larger (smaller) the absolute value of E_d, the more (less) responsive the quantity demanded to a price change.

Measuring price elasticity of demand

- The **arc elasticity of demand** is used to measure the price elasticity of demand over a certain price range of a demand curve.

$$E_d = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

$$= \frac{\frac{Q_{d1} - Q_{d0}}{(Q_{d0} + Q_{d1})/2} \times 100\%}{\frac{P_1 - P_0}{(P_0 + P_1)/2} \times 100\%}$$

$$= \frac{Q_{d1} - Q_{d0}}{(Q_{d0} + Q_{d1})/2} \times \frac{(P_0 + P_1)/2}{P_1 - P_0}$$

$$= \frac{Q_{d1} - Q_{d0}}{Q_{d0} + Q_{d1}} \times \frac{P_0 + P_1}{P_1 - P_0}$$

Let
P_0 = Original price
Q_{d0} = Original quantity demanded
P_1 = New price
Q_{d1} = New quantity demanded

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Types of price elasticity of demand

Type of price elasticity of demand	Numerical value of E_d	Features
Inelastic	$0 < E_d < 1$	**The percentage change in quantity demanded of the good is smaller than the percentage change in price.**
Elastic	$1 < E_d < \infty$	**The percentage change in quantity demanded of the good is greater than the percentage change in price.**
Unitarily elastic	$E_d = 1$	▪ **The percentage change in quantity demanded of the good is equal to the percentage change in price.** ▪ The demand curve can be represented by a **rectangular hyperbola**.
Perfectly inelastic	$E_d = 0$	▪ **A change in price does not cause any change in the quantity demanded.** ▪ The demand curve can be represented by a **vertical line**.
Perfectly elastic	$E_d = \infty$	▪ **An extremely small change in price causes an infinite change in quantity demanded.** ▪ The demand curve can be represented by a **horizontal line**.

- A less elastic demand is not necessarily an inelastic demand unless the absolute value of E_d drops below 1.
- A demand curve that has a steep slope and/or cuts the x-axis may not be inelastic. The value of E_d has to be calculated.

10.2 Factors Affecting Price Elasticity of Demand

- Availability and attractiveness of substitutes: The greater (smaller) the number of attractive substitutes available in the market, the more (less) elastic the demand for a good. The substitutability of a good can be affected by:
  - how necessary a good is to consumers goods will increase.
  - whether people are addicted to a good
  - how a good is categorised
- Proportion of income spent on a good: The smaller (bigger) the proportion of income spent on a good, the less (more) elastic the demand for the good.
- Time for adjusting consumption patterns: The shorter (longer) the time for adjusting consumption patterns, the less (more) elastic the demand for the good.
- Number of uses: The demand for a good with more (fewer) uses tends to be more (less) elastic.
- Durability: The demand for durable goods tends to be more elastic.
- Price ranges: For a downward-sloping liner demand curve, the higher (lower) the price range, the more (less) elastic the demand is.

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10.3 Relationship between Price Elasticity of Demand and Total Revenue

Meaning of total revenue

$$\text{Total revenue (TR)} = \text{Price (P)} \times \text{Quantity transacted (Q)}$$

Total revenue (sellers) = Total expenditure (buyers) = Total market value (society)

Effects of a change in price on total revenue

- An increase in price
  → gain in revenue from selling the goods with a higher price
  → loss in revenue with lower quantity transacted
- A decrease in price
  → gain in revenue with higher quantity transacted
  → loss in revenue from selling the goods with a lower price

Different types of price elasticity of demand and total revenue

Type of price elasticity of demand	Effect of changes in price on total revenue (TR)	
	When price increases	When price decreases
Inelastic ($0 < E_d < 1$)	• % decrease in Q_d < % increase in P • loss in TR < gain in TR • TR increases	• % increase in Q_d < % decrease in P • gain in TR < loss in TR • TR decreases
Elastic ($1 < E_d < \infty$)	• % decrease in Q_d > % increase in P • loss in TR > gain in TR • TR decreases	• % increase in Q_d > % decrease in P • gain in TR > loss in TR • TR increases
Unitarily elastic ($E_d = 1$)	• % decrease in Q_d = % increase in P • loss in TR = gain in TR • TR remains unchanged	• % increase in Q_d = % decrease in P • gain in TR = loss in TR • TR remains unchanged
Perfectly inelastic ($E_d = 0$)	• Q_d remains unchanged • TR increases	• Q_d remains unchanged • TR decreases
Perfectly elastic ($E_d = \infty$)	• Q_d drops to zero • TR drops to zero	• Q_d drops to zero • TR drops to zero

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10.4 Price Elasticity of Supply

Meaning of price elasticity of supply

- **Price elasticity of supply** measures how responsive the quantity supplied of a good is to a change in its price.

$$\text{Price elasticity of supply coefficient } (E_S) = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

- According to the law of supply, the price elasticity of supply coefficient (E_S) is always positive.

Measuring price elasticity of supply

- The **arc elasticity of supply** is used to measure the price elasticity of supply over a certain price range of a supply curve.

$$E_S = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

$$= \frac{\frac{Q_{S1} - Q_{S0}}{(Q_{S0} + Q_{S1})/2} \times 100\%}{\frac{P_1 - P_0}{(P_0 + P_1)/2} \times 100\%}$$

$$= \frac{Q_{S1} - Q_{S0}}{(Q_{S0} + Q_{S1})/2} \times \frac{(P_0 + P_1)/2}{P_1 - P_0}$$

$$= \frac{Q_{S1} - Q_{S0}}{Q_{S0} + Q_{S1}} \times \frac{P_0 + P_1}{P_1 - P_0}$$

Let
P_0 = Original price
Q_{S0} = Original quantity supplied
P_1 = New price
Q_{S1} = New quantity supplied

Types of price elasticity of supply

Type of price elasticity of supply	Numerical value of E_s	Features
Inelastic	$0 < E_s < 1$	**The percentage change in quantity supplied is smaller than the percentage change in price**.
Elastic	$1 < E_s < \infty$	**The percentage change in quantity supplied is greater than the percentage change in price**.
Unitarily elastic	$E_s = 1$	▪ **The percentage change in quantity supplied is equal to the percentage change in price**. ▪ The supply curve can be represented by any upward-sloping straight line passing through the origin.
Perfectly inelastic	$E_s = 0$	▪ **A change in price does not cause any changes in the quantity supplied**. ▪ The supply curve can be represented by **a vertical line**.
Perfectly elastic	$E_s = \infty$	▪ **An extremely small change in price causes an infinite change in quantity supplied**. ▪ The supply curve can be represented by **a horizontal line**.

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10.5 Factors Affecting Price Elasticity of Supply

- Flexibility in adjusting production levels: Higher (lower) flexibility in adjusting production levels leads to a more (less) elastic supply. The flexibility is affected by:
  - spare production capacity
  - availability of critical factor inputs
  - nature of the industry or product
- Time for adjusting production levels: When the time for adjusting production levels is shorter (longer), the supply of a good tends to be less (more) elastic.
- Ease of entry of new firms: When there are more (fewer) entry barriers to an industry, the supply of the good of the industry tends to be less (more) elastic.

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